

PROJECT JOURNAL

TOPIC: Cross-Language Personalization for Dataset Selection Using BAAI General Embedding

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WEEK 1

Date : 5/12/24

Task : Dataset selection

For the first week I have spend my time for the dataset selection which is very crucial for our project. So here I have selected a good Malayalam dataset, my focus was on creating the infrastructure and mechanisms necessary to facilitate the interaction of the chatbot in the Malayalam language. I really wanted to make it seamless for the Malayalam speaking users to use the chatbot. so with the limitation as a regional language I really need to select a good one for this. After comparing many datasets I have chosen the Dakshina dataset for doing my part. For that:

- Analyzed the Dakshina dataset to understand the structure of Malayalam language data.
- Cleaned the dataset by converting Romanized and native Malayalam text into a consistent format.
- Identified key columns in the dataset, such as transliteration pairs

Time Spent: 8 hours

Challenges Faced:

- Dataset limitations due to the language limitations
- Difficulty in handling mixed Romanized Malayalam with native Malayalam script.
- Ensuring proper mapping of transliterated words to native words without losing context.

Solutions:

- Validated the dataset by cross-checking the transformations with a sample of known words.
- Check if there is any misunderstanding in the meaning by crosschecking with the English content.

Learning Outcomes:

- Gained experience in preprocessing multilingual text data and handling different writing systems.
- Learned how to handle issues in text normalization for natural language processing (NLP).

WEEK 2

Date : 12/12/24

Task : Data Analysis

In the second week , I performed some EDA on the Dakshina dataset-a collection of text data in Malayalam. My goal was to understand the pattern of language usage from the frequency distributions of words and phrases in Malayalam through various document types. These have been divided into broad categories: formal and informal content. I focused on the visualization of these distributions using various techniques to identify common patterns, trends, and linguistic features in the data. This included:

- Through the word frequency analysis I analysed how frequently certain words and phrases appeared in the dataset to identify the most commonly used words in Malayalam across different contexts.
- I have also done the metadata analysis by using the data available in the dataset, such as sentence lengths, average word counts, and usage statistics.

Time Spent: 6 hours

Challenges Faced:

- The out-of-vocabulary or rare words will make it hard to understand; sometimes there are many instances of domain-specific or rare words not appearing frequently enough in the dataset.
- These rare words caused inconsistencies to occur in the frequency analysis, which makes it difficult to get a proper representation of general language trends.
- • Another point is the differentiation between formal and informal. It wasn't clear from the dataset as labels, and also in the formal documents, some of the instances contained informal phrases or colloquial expressions

Solutions:

- I applied various filtering techniques to remove low-frequency words that were contributing to the noise in the analysis.
- In order to deal with the issue of formal and informal, I introduced a context-based classification approach. It was done by checking the sentence structure, word choice, and stylistic elements in the dataset to decide whether the document was formal or informal.

Learning Outcomes:

- I improved the robustness of handling outliers and rare words in large language datasets.
- I developed a better understanding of the complexity associated with classifying language data into various contexts.
- Working on metadata and looking at sentence structures gave me a greater appreciation for Malayalam syntax, including word order and variation in sentence length when comparing formal and informal contexts.

WEEK 3

Date : 20/12/24

Task : Embedding

This week I have focused on the tokenizing and text embedding techniques to be performed on the Malayalam dataset. It is one of the most crucial steps in data preparation for most NLP applications that my chatbot stands on. The objective was to convert raw Malayalam text into machine-readable vector representations that capture semantic and syntactic nuances, enabling the chatbot to process and respond intelligently to Malayalam inputs.

- First is the tokenization process in that I have breakdown the text into smaller units such as words or subwords, while preserving linguistic meaning. And it requires the special attention because we need to ensure that the meaning of complex words was retained.
- I used BGE, which is a multi-purpose model for generating vector representations of text. These embeddings were meant to encode Malayalam text into vector forms retaining the contextual and semantic information of the original input. These are crucial in the downstream tasks of intent recognition and response generation in the NLP engine of the chatbot.
- I placed emphasis on optimizing the efficiency of text encoding.

Time Spent: 9 hours

Challenges Faced:

- The tokenization challenges are that the most words include complex ligatures and conjunct characters, which make it difficult for many existing libraries to segment text properly.
- Mis-tokenization could result in a loss of meaning and affect subsequent embedding generation and NLP tasks.
- Capturing the nuances of Malayalam syntax and vocabulary, which often differs significantly from widely studied languages like English.

Solutions:

To address these challenges, I implemented the following solutions:

- Uses the INDIC-NLP library which is specifically designed to handle Indian languages including Malayalam. This adaptation of the library's tokenization functionality let me achieve to tokenize the Malayalam text into meaningful tokens without losing linguistic integrity.
- Applied the BGE (BAAI General Embeddings) model, which provided a strong foundation for creating vector representations.
- Evaluating embedding performance on test cases, such as identifying the similarity between pairs of sentences or words.

Learning Outcomes:

- I acquired much knowledge on handling complex tokenization challenges for Malayalam, such as the importance of preserving meaning in agglutinative languages.
- Working with BGE embeddings helped me to understand how syntax and semantics are combined to encode the text.

WEEK 4

Date : 27/12/24

Task : Chatbot

In the final week of this project, I worked on the development and testing of the chatbot based on the pre-processed Malayalam dataset. The variety of user inputs was handled through queries in Malayalam and mixed Malayalam-English. The tasks that I did are:

- Incorporate a language model that can understand natural language and generate text in Malayalam.
- Optimizing the architecture of the chatbot for conversational contexts, so it could provide clear and contextually relevant responses.
- I analysed the performance metrics of the deployed chatbot for Response Accuracy and Error Handling and found instances where either the chatbot had failed to respond correctly or had misinterpreted the query.

Time Spent: 8 hours

Challenges Faced:

- Processing the transliterated text is really complicated because the Standard NLP tools struggled to handle such transliterated text.
- The fine-tuning of the chatbot language model to handle mixed language and transliterated queries was extremely computational- and time-intensive. Ensuring that the chatbot maintained high accuracy across all query types required numerous iterations.

Solutions:

- Used a hybrid translation model to handle mixed-language inputs, incorporating both Malayalam and English responses.
- Fine-tuning on real-world query data to reach high accuracy for question understanding and response.

Learning Outcomes:

- Enhanced my skills in integrating the models with multilingual datasets, specifically for Malayalam.
- Gained valuable insights into handling mixed-language datasets in chatbot applications.

SUMMARY

Total time spend : 30+ hours

Key Accomplishments:

- Processed and cleaned the Dakshina dataset to ensure consistent and meaningful text data.
- Conducted detailed exploratory data analysis to identify key language usage patterns.
- Developed text embeddings using BGE and integrated them into the chatbot's NLP engine.
- Tested and improved the chatbot's performance in Malayalam and mixed-language queries.